

Why Context Matters?

Retrieval-Augmented Generation



Good Context = Good Results



Does the new 'Aether-5' drone have
4K video recording?

User

The Aether drone series is known for its excellent
camera quality. Most modern drones in its class
support 4K video, so it is highly likely that the Aether-5
does as well.

Chatbot



Does the new 'Aether-5' drone have
4K video recording?

User

Context: The Aether-5 drone, launched in September 2025, features a new 1-inch sensor. This enables video capture in 6K at 30fps and 4K at up to 120fps, a significant upgrade from the Aether-4's 1080p capability

Yes, the Aether-5 drone supports 4K video recording at
up to 120fps. It can also record in 6K at 30fps.

Chatbot

Context Window: The Model's Memory



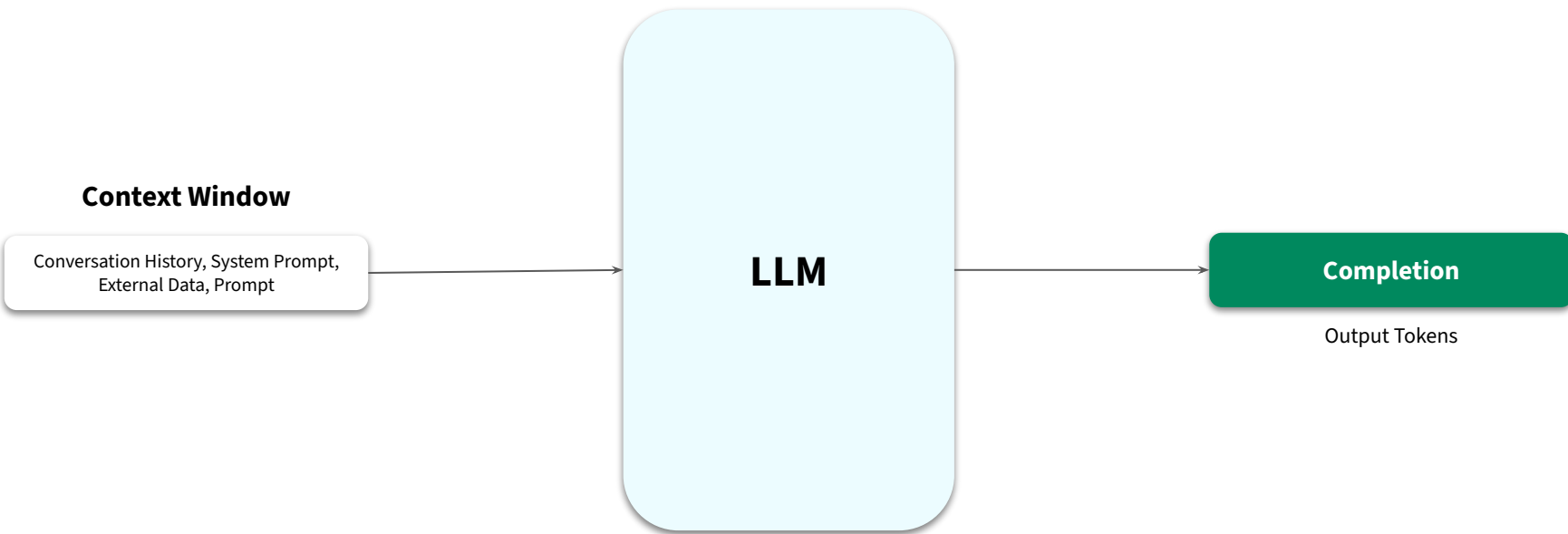
Context Window

Conversation History, System Prompt,
External Data, Prompt

LLM

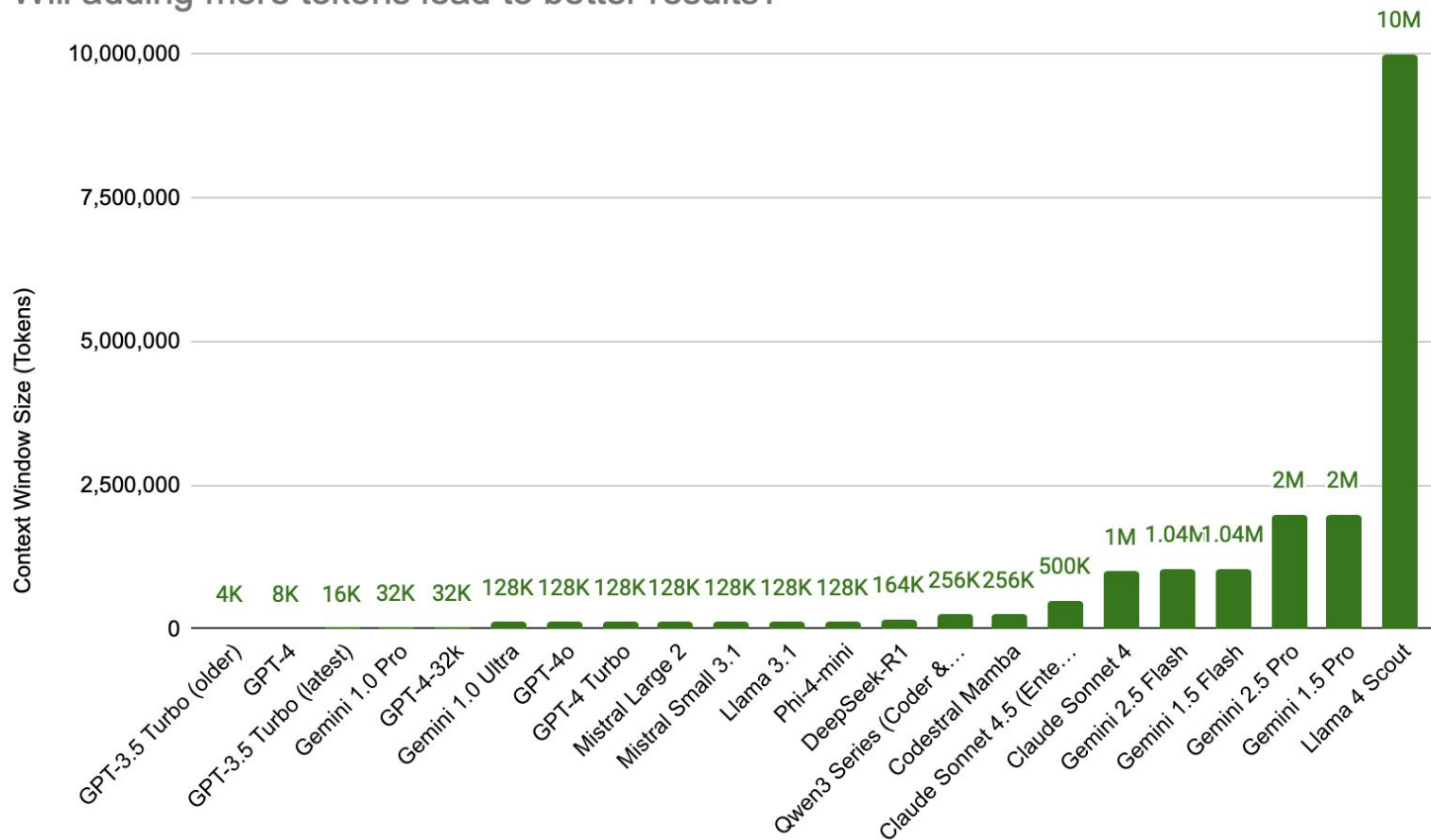
Completion

Output Tokens





Will adding more tokens lead to better results?





No.



Needle in the Haystack







Good, Small Context = Better Results

When to use RAG

Retrieval-Augmented Generation



Business Factors

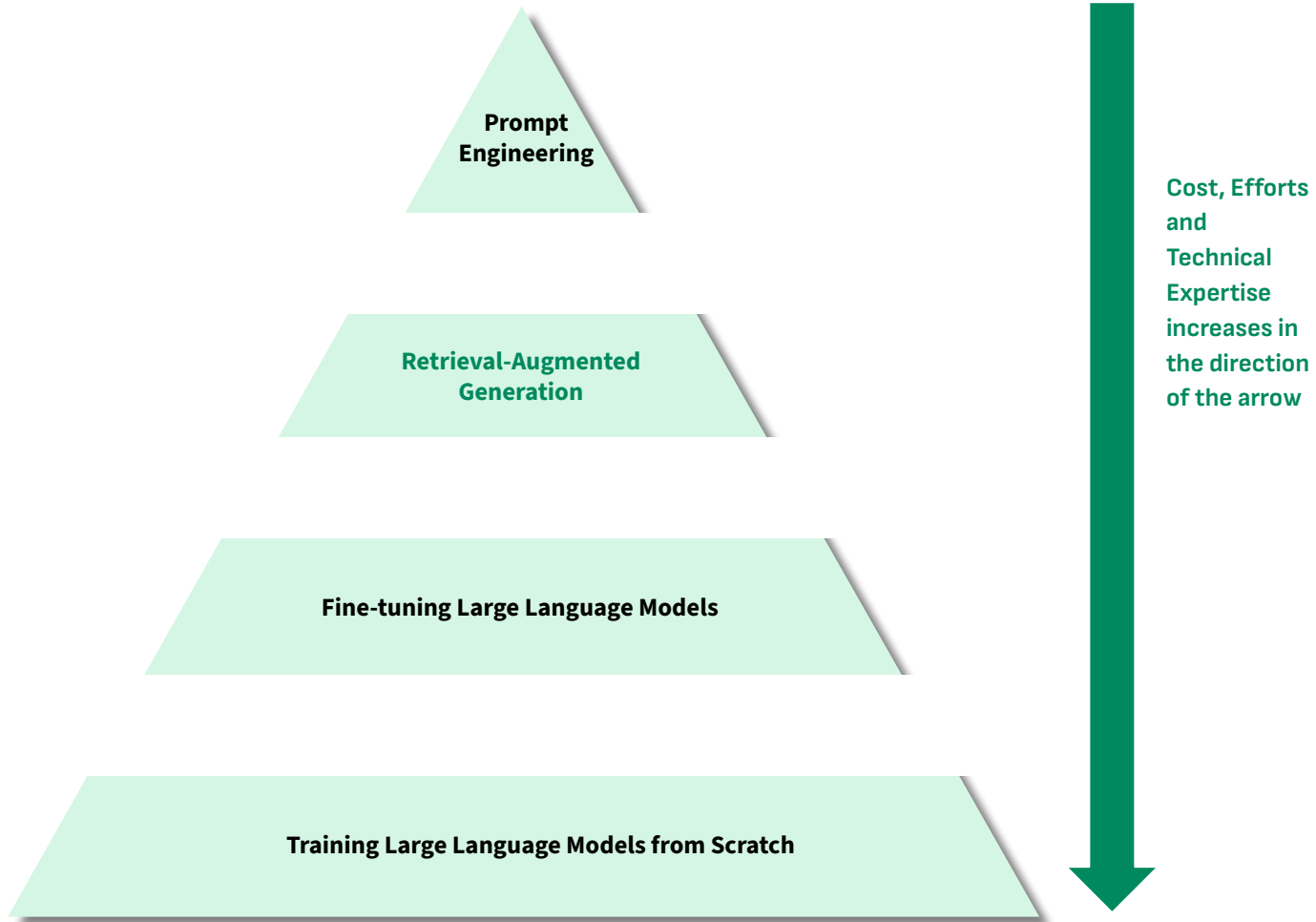


Business Factors

Cost

Efforts

Technical Expertise

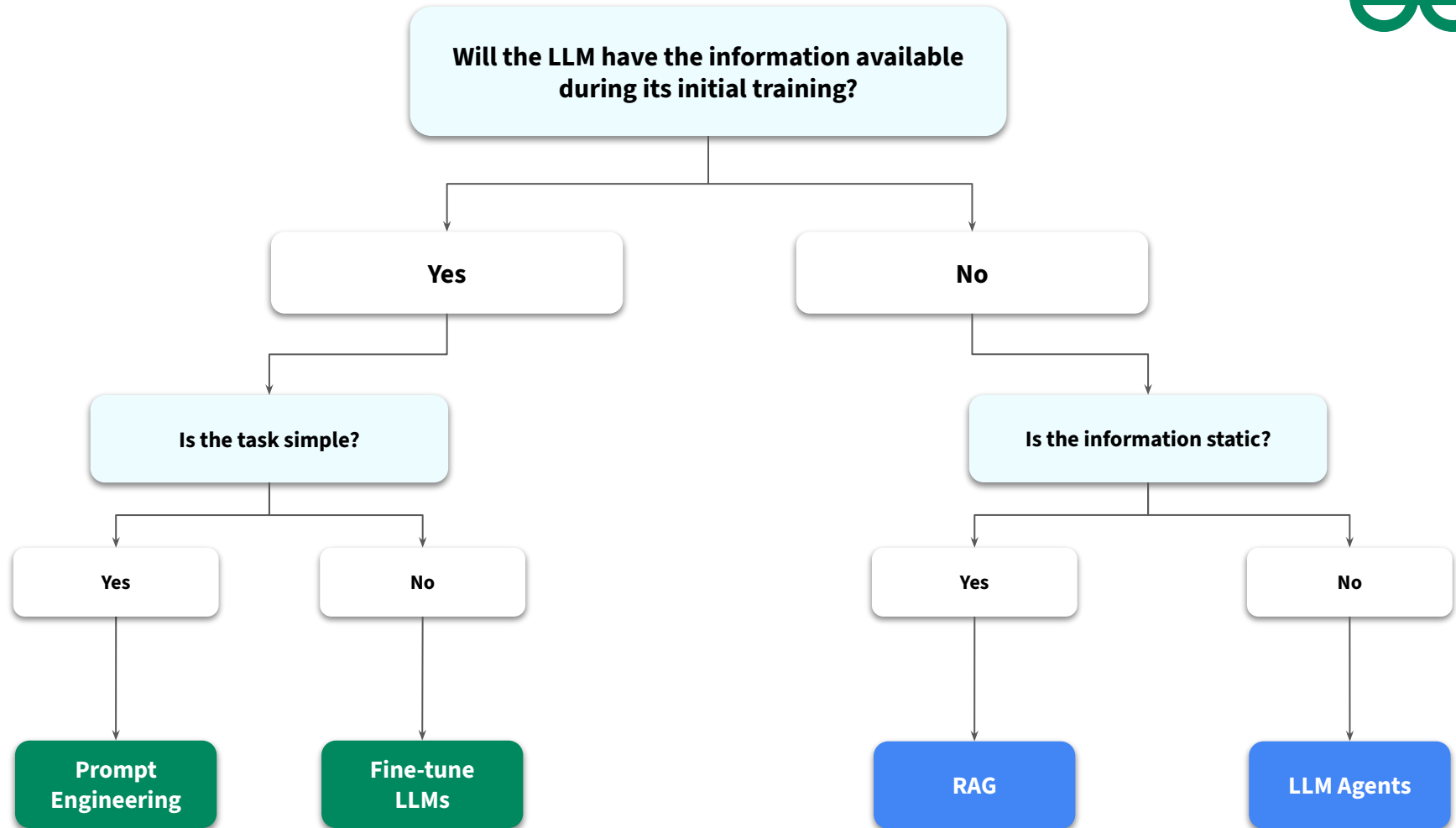




Technical Perspective



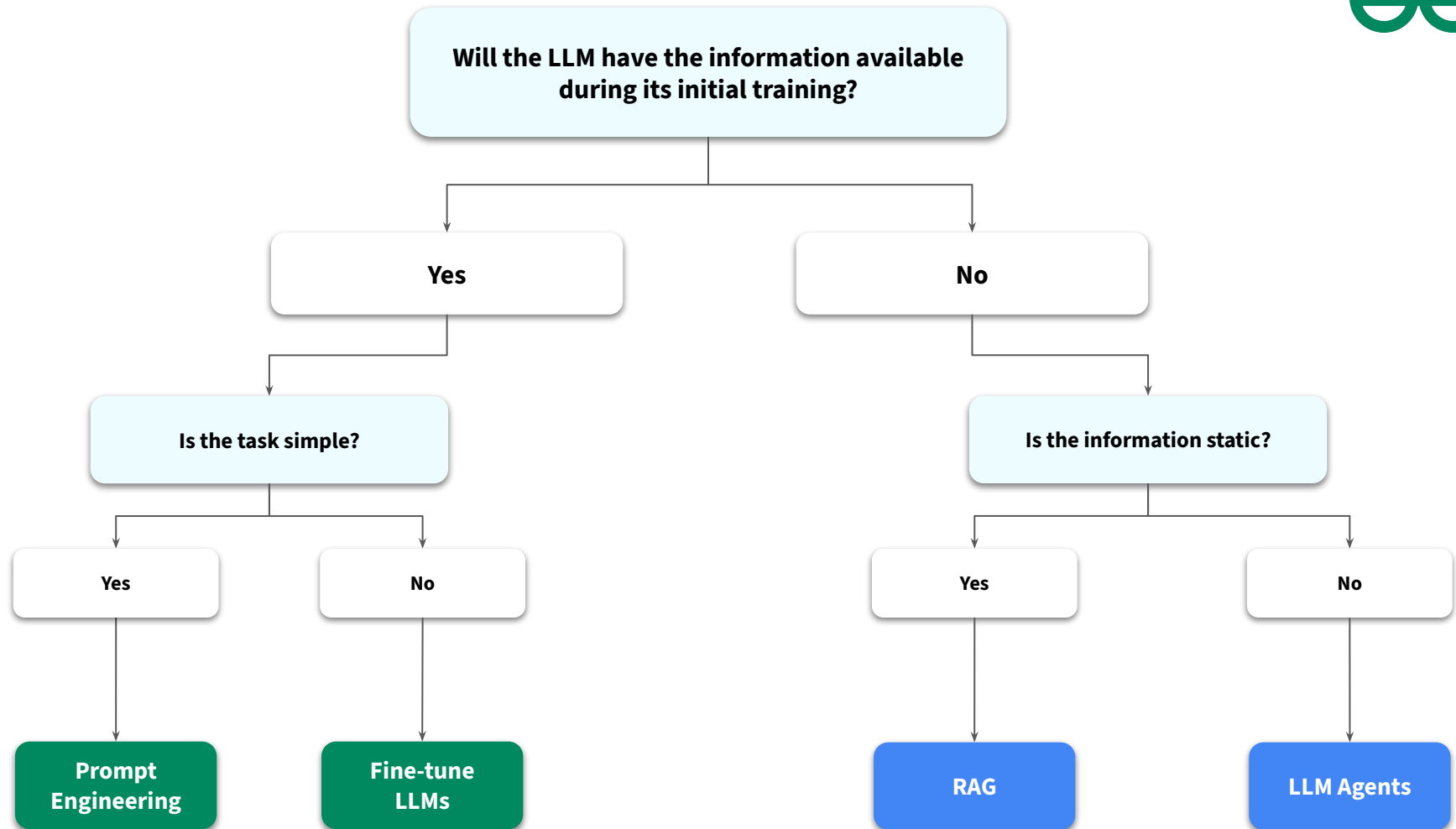
Information Required for the Task

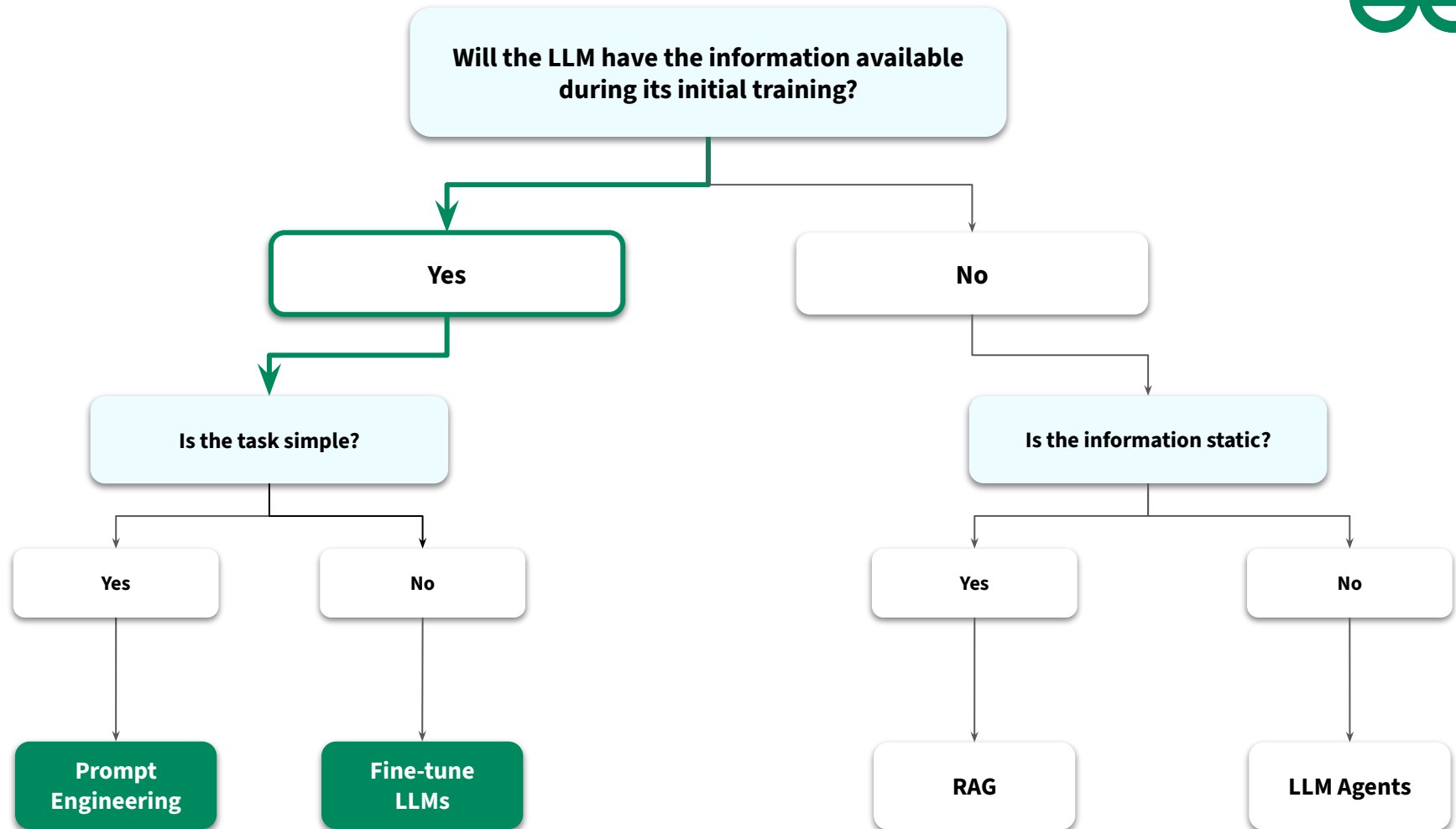


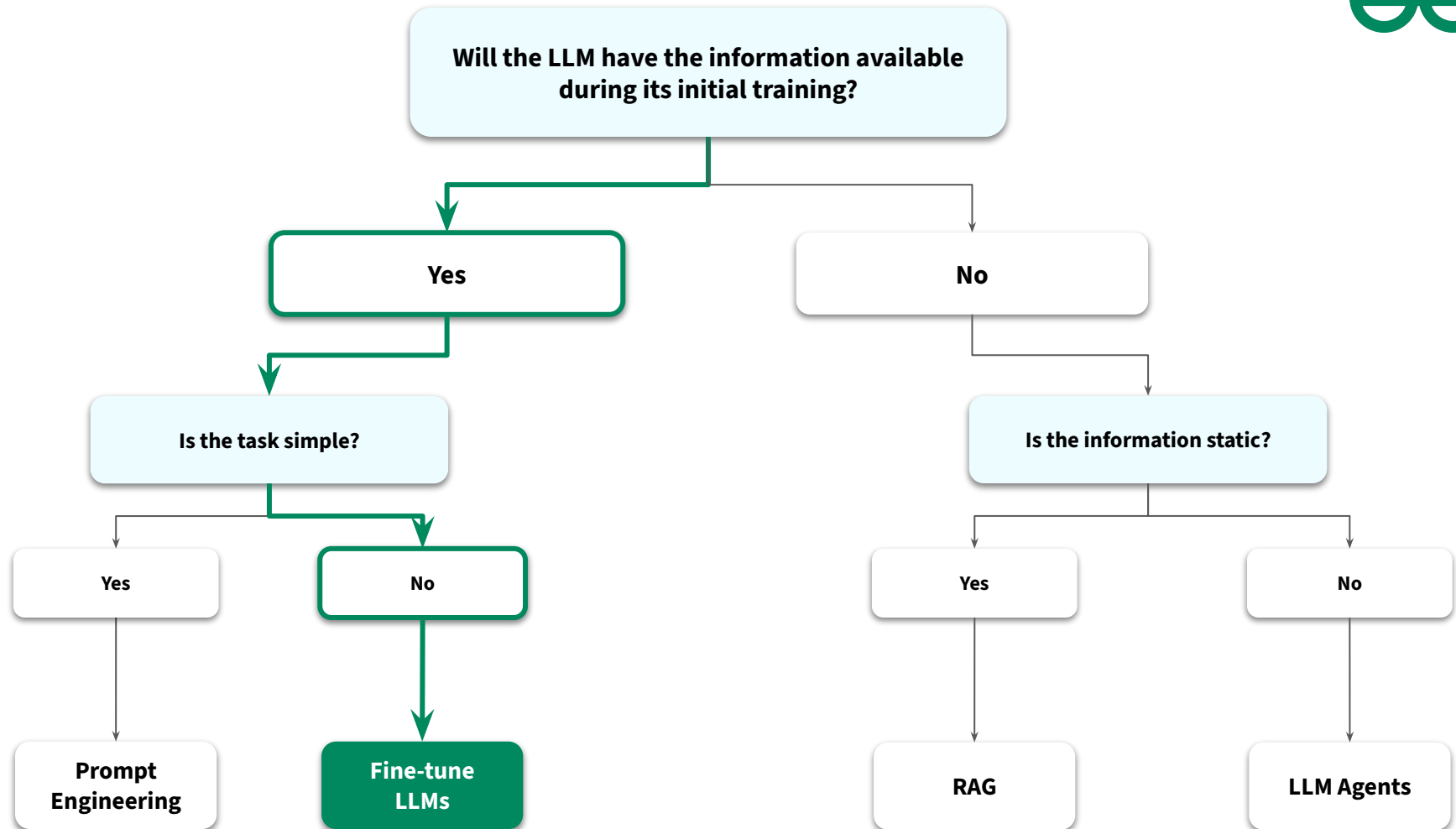


Scenario One

You want to use LLMs to generate medical diagnosis using medical records



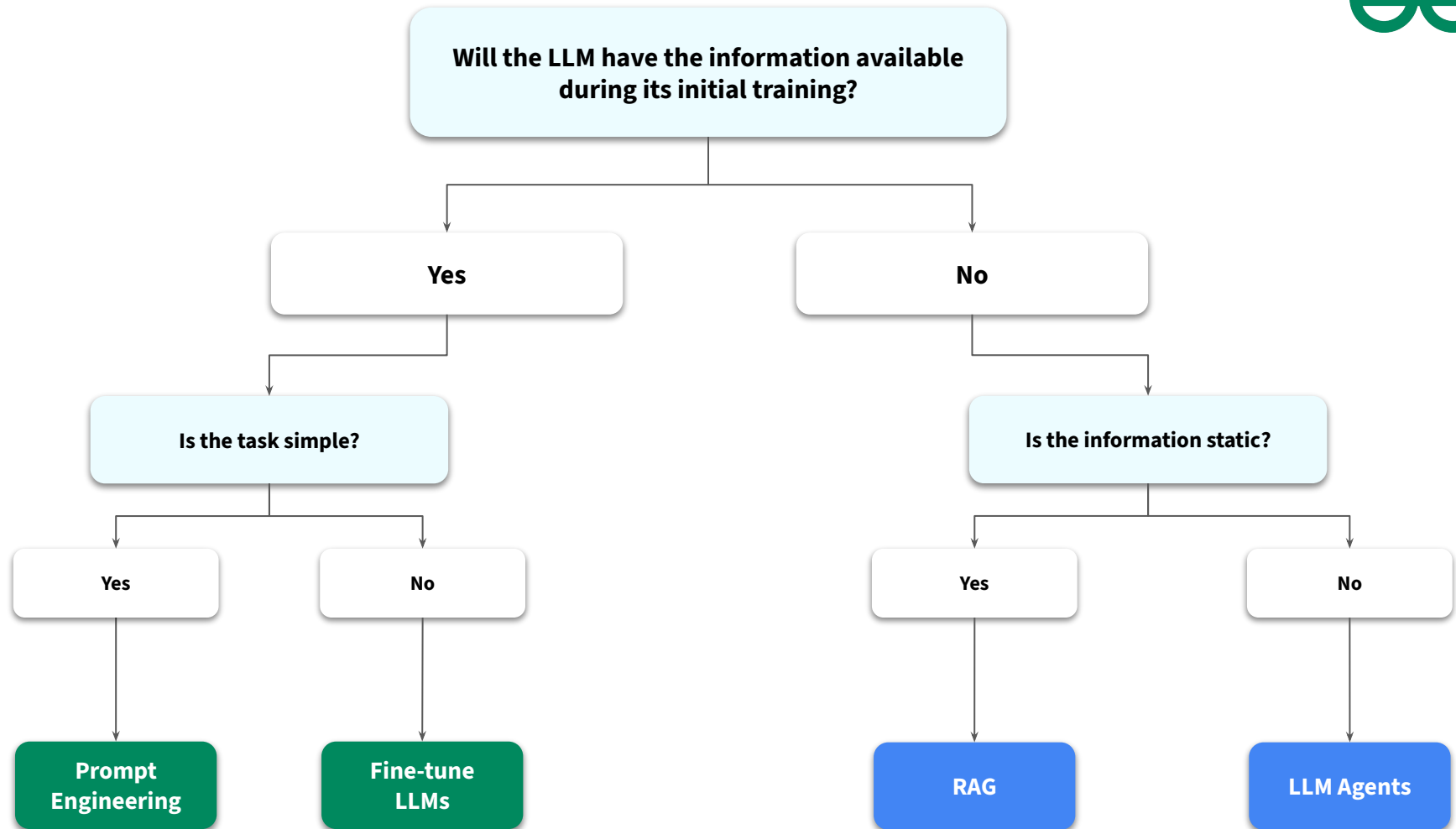


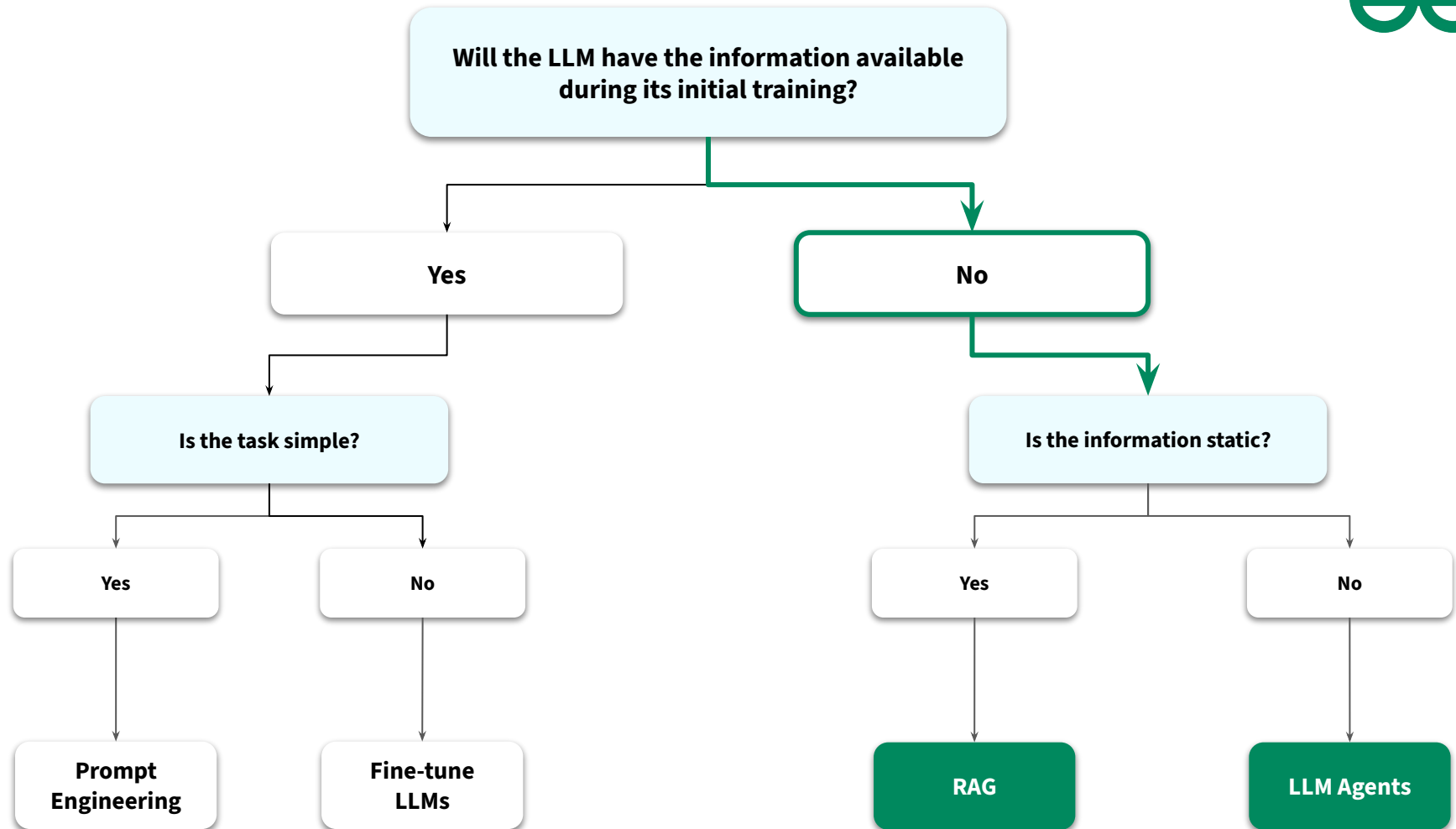


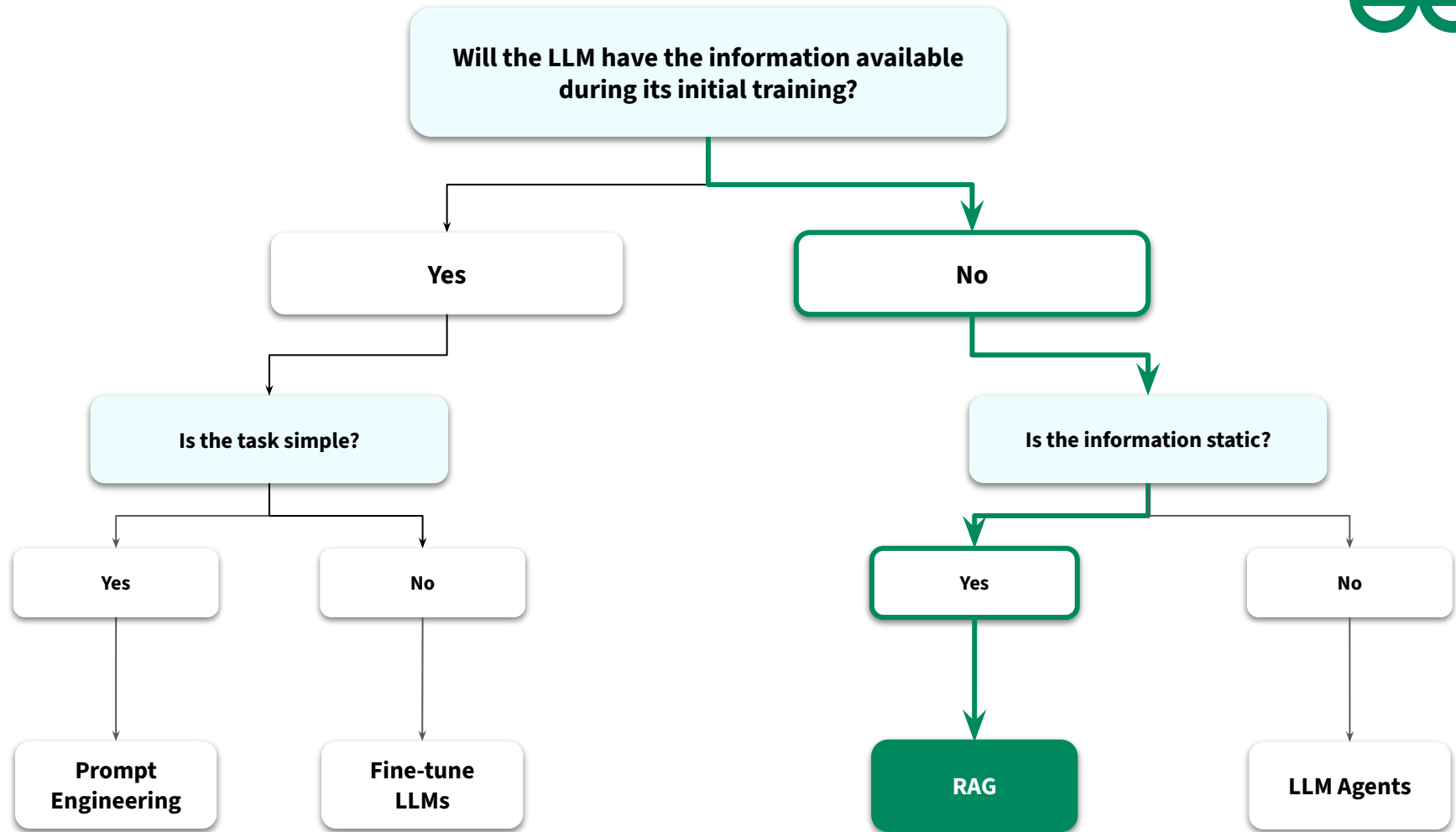


Scenario Two

You want to develop a chatbot that helps your team access all company policy documents and answers any related questions.





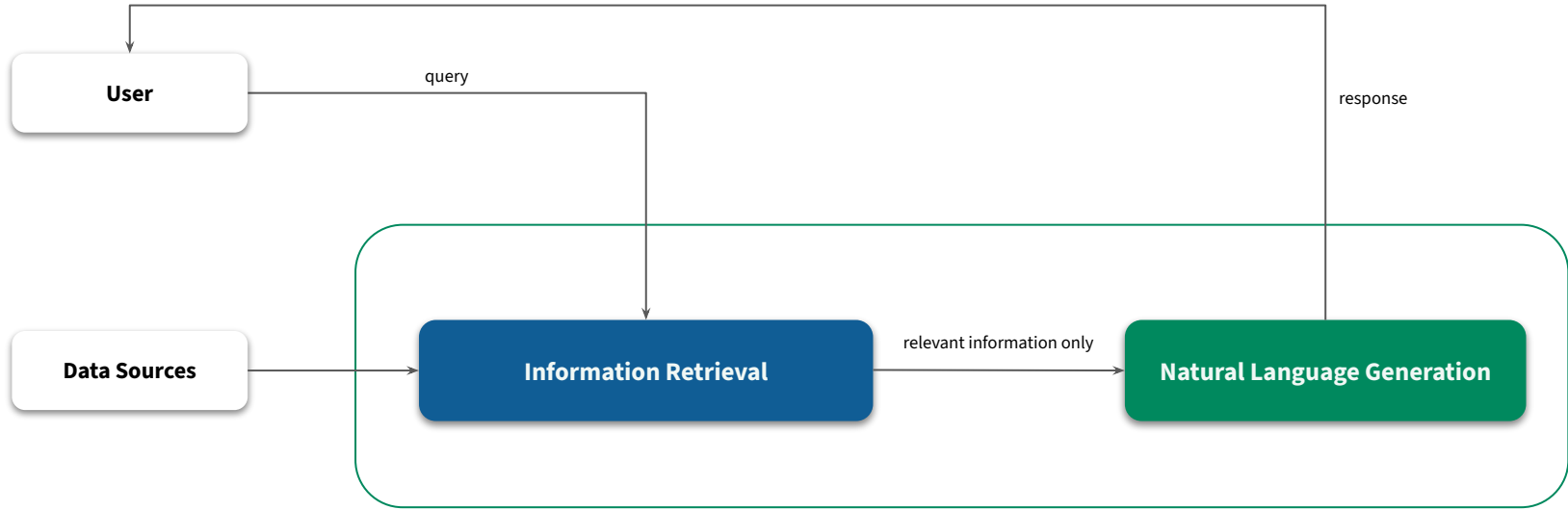


RAG Deep Dive

Retrieval-Augmented Generation

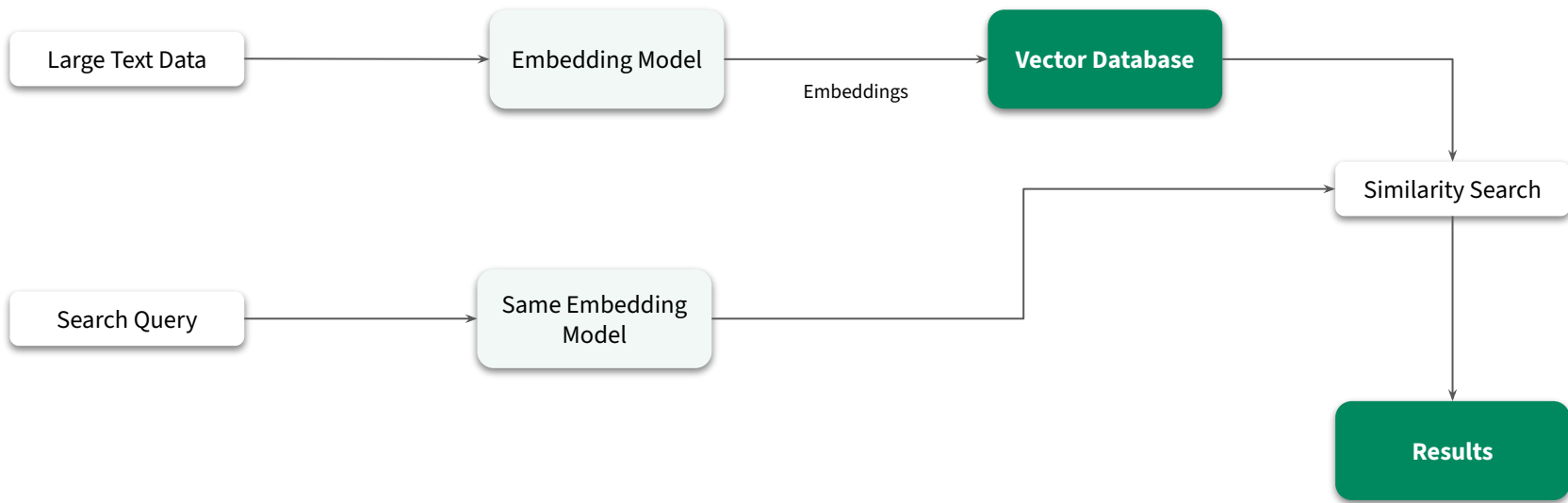


RAG

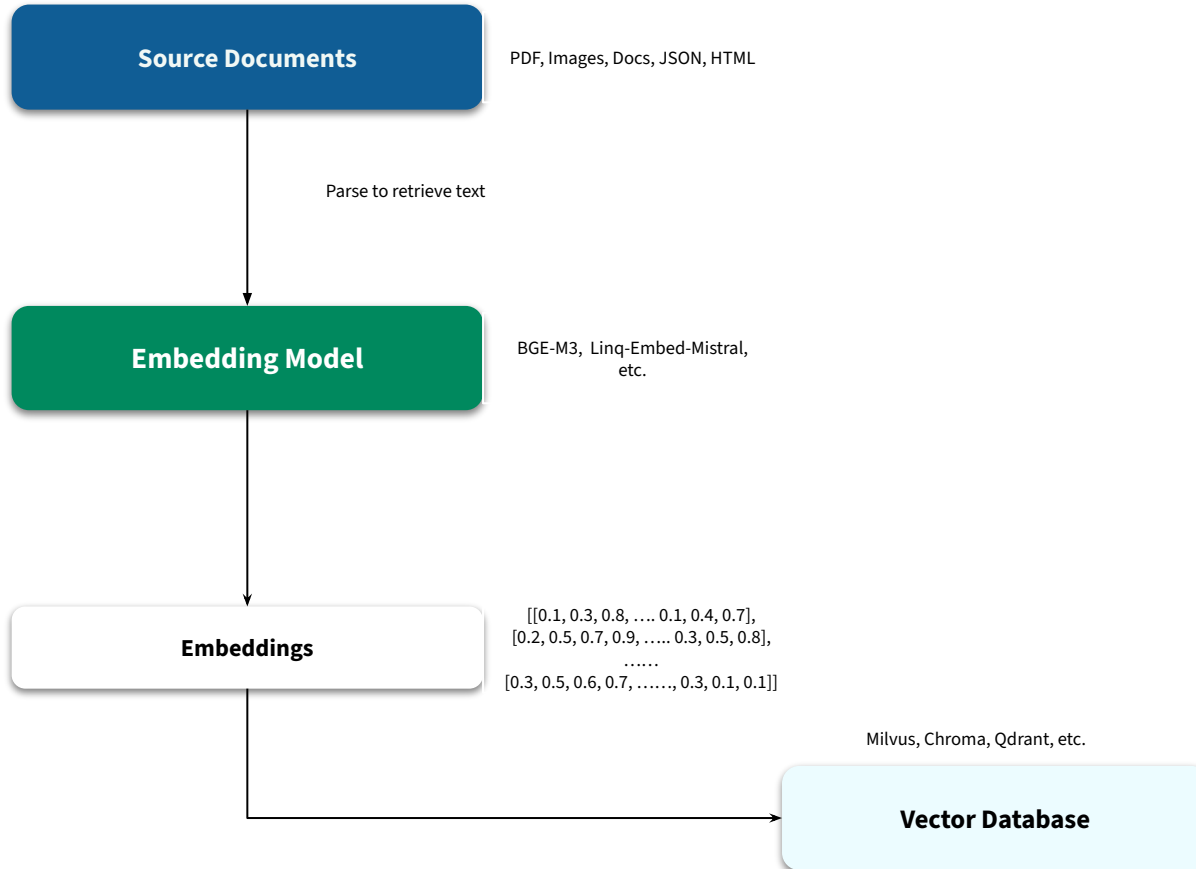




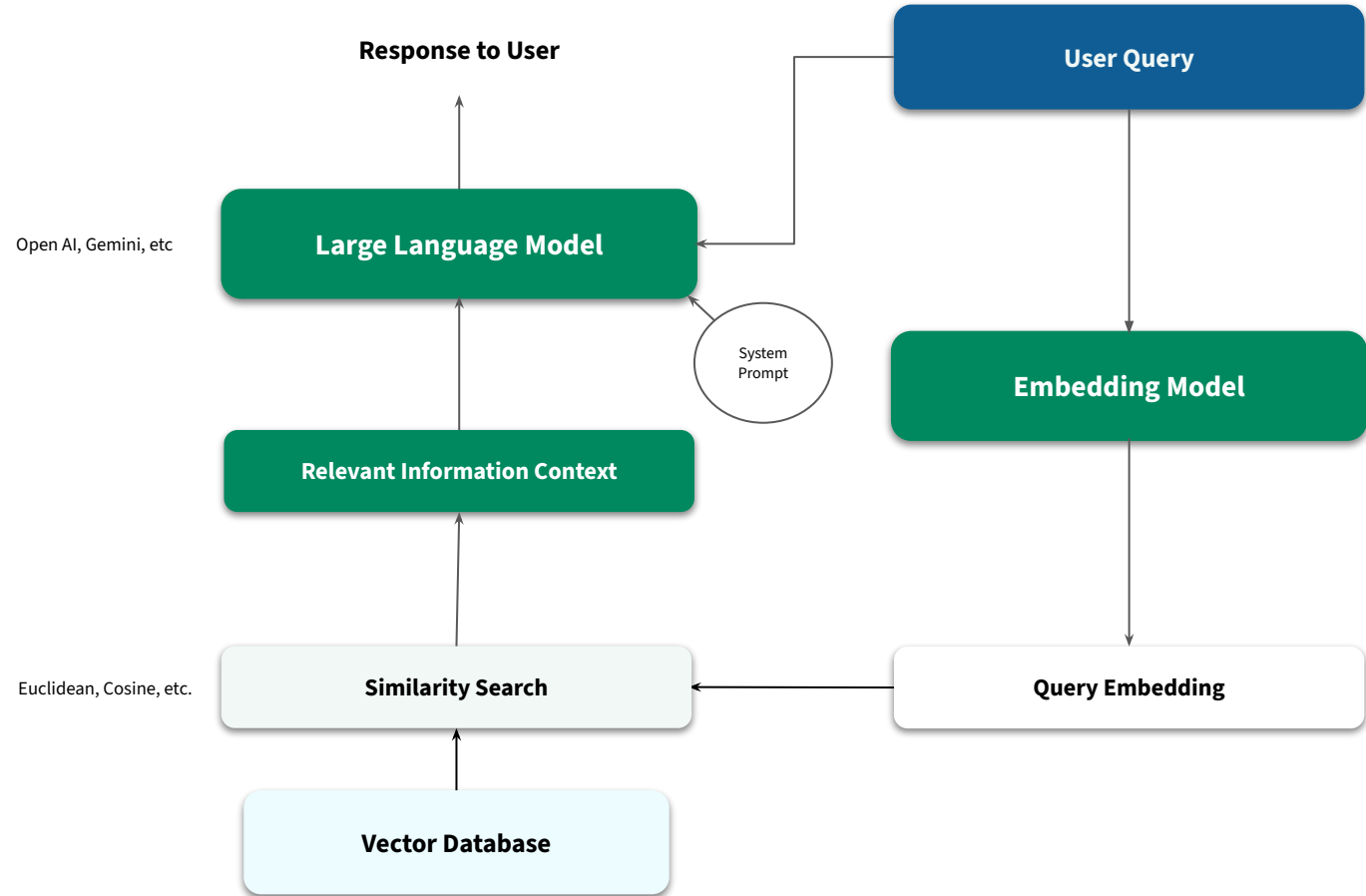
Tangent: Remember Vector Databases?



Information Retrieval: The Quest for Relevance



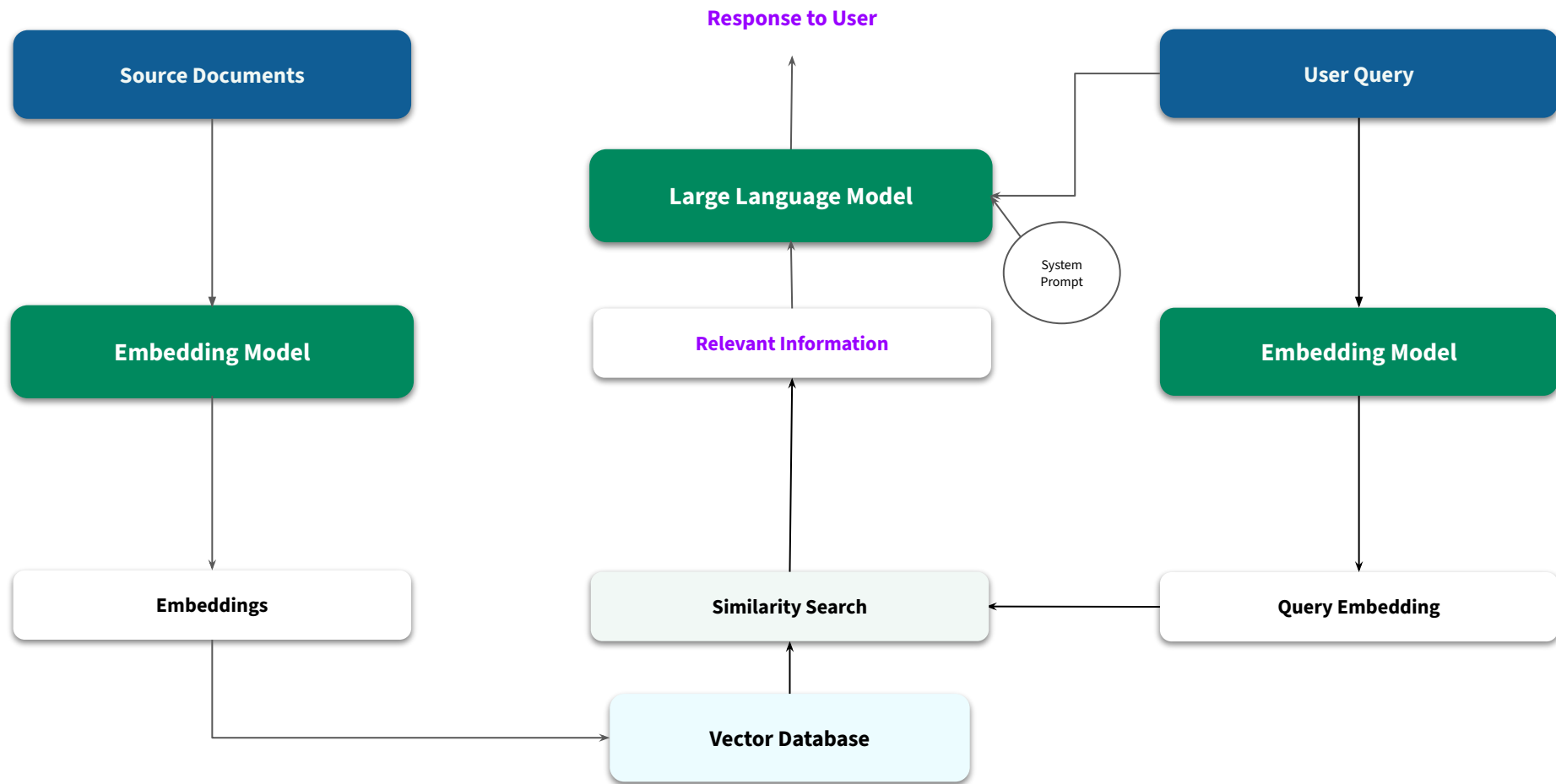
Natural Language Generation: The Art of Articulation





1. If embeddings are similar, original sentences are similar and have similar meanings.
2. Vector database will contain only answer-based sentences (which are stored as embeddings).
3. If query embedding is similar to one of the embeddings in the vector database, then the two sentences are similar.
4. But, one is a question and the other is an answer.

RAG Flow



Chunking

Retrieval-Augmented Generation



So, what is Chunking?



Fixed-Size Chunking



., ; :-

Recursive-Based Chunking



Semantic Chunking



Contextual Chunking with LLMs

Challenges

Retrieval-Augmented Generation



RAG is not a perfect solution

Managing Multiple Data Sources

Large Volume Data

Incremental Updates

Scaling the Retrieval Mechanism

Historical & Ethical Considerations

Implementation with LangChain

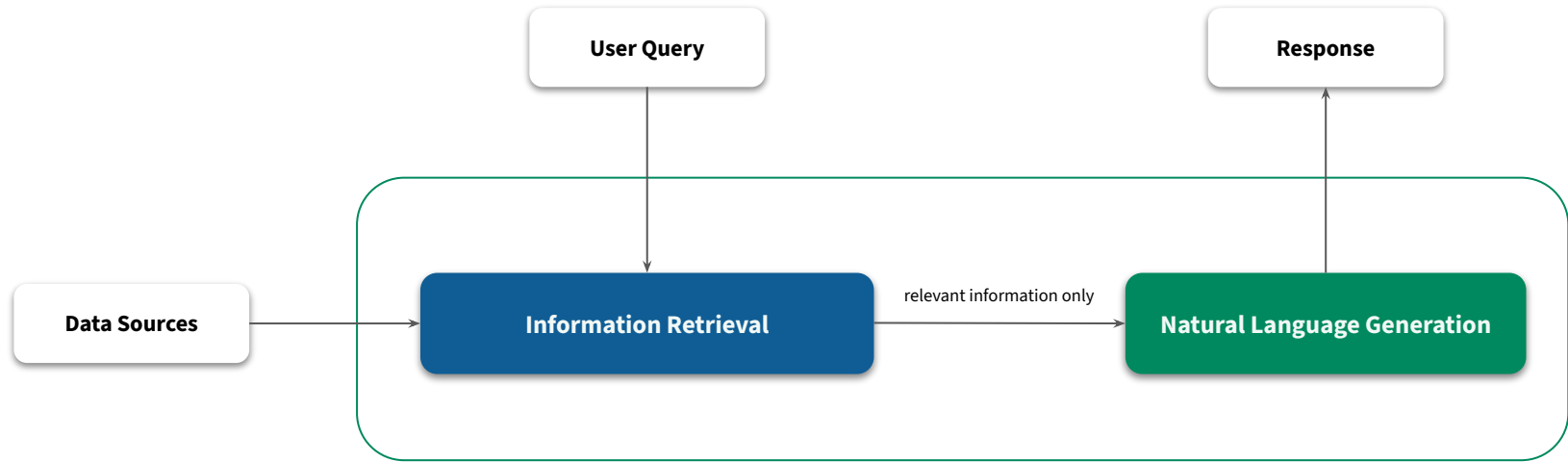
Retrieval-Augmented Generation

Variations

Retrieval-Augmented Generation

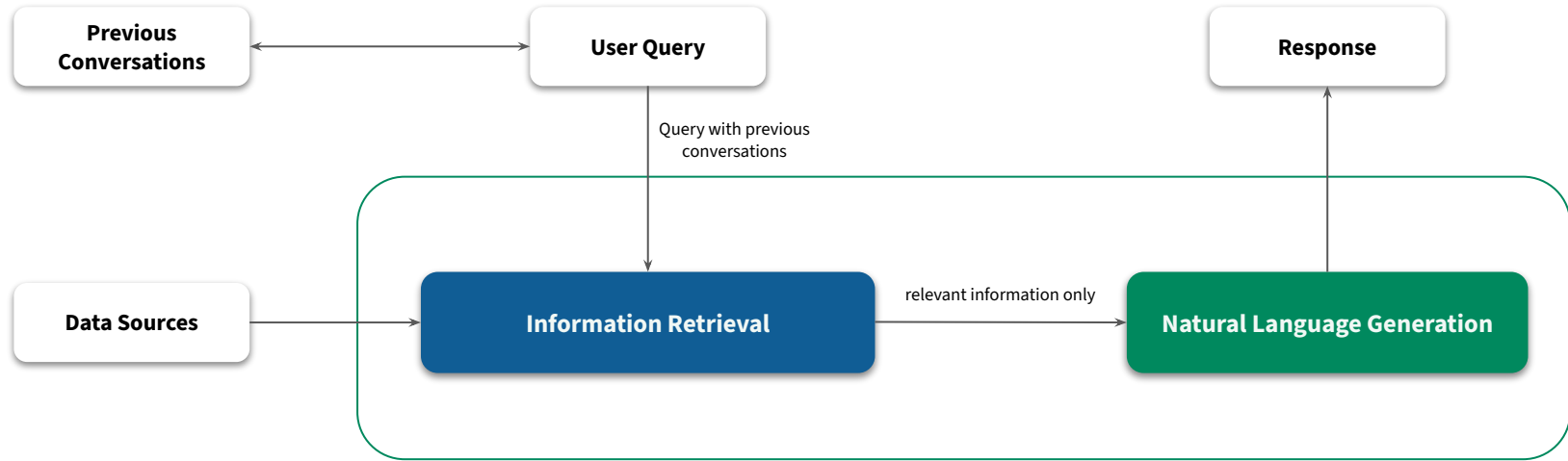


Simple or Naive RAG



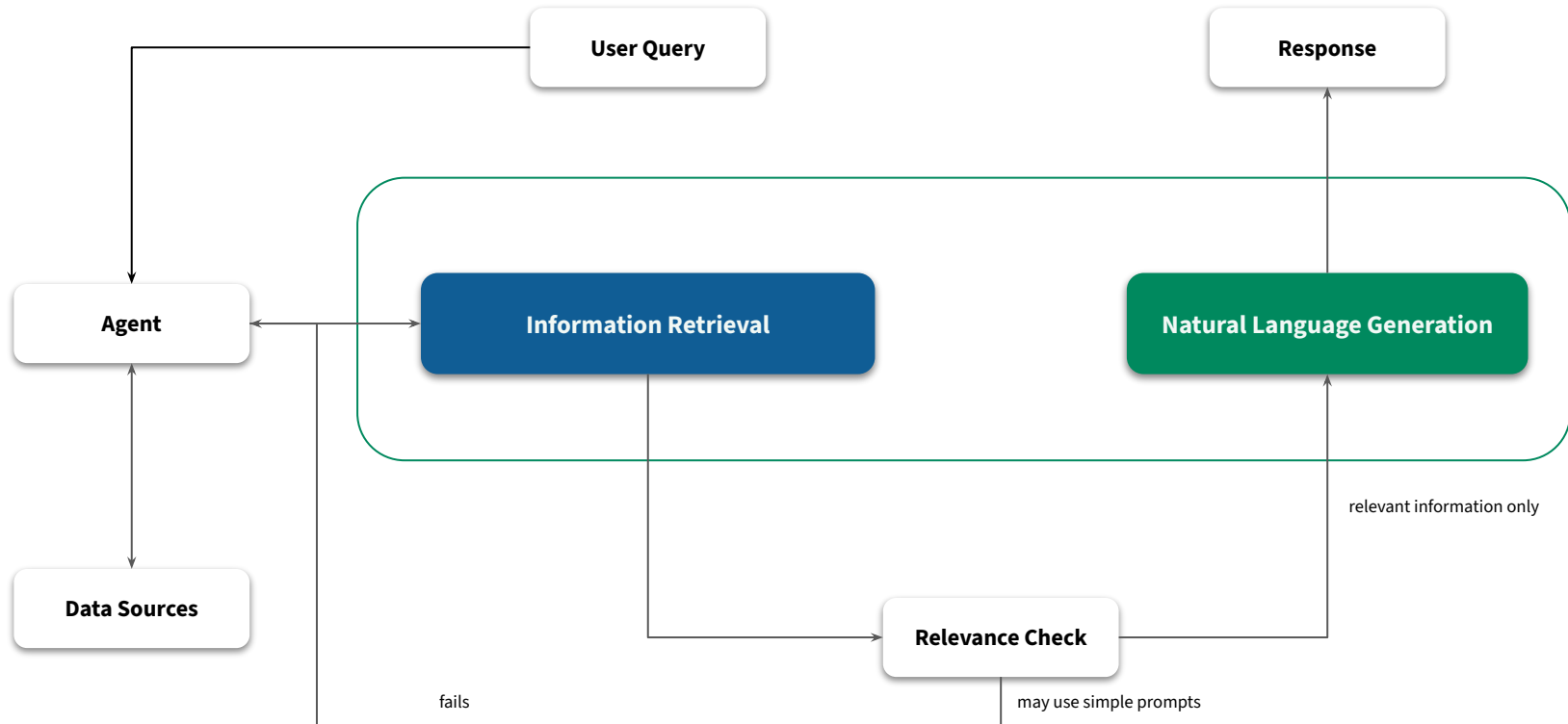


Simple RAG with Memory



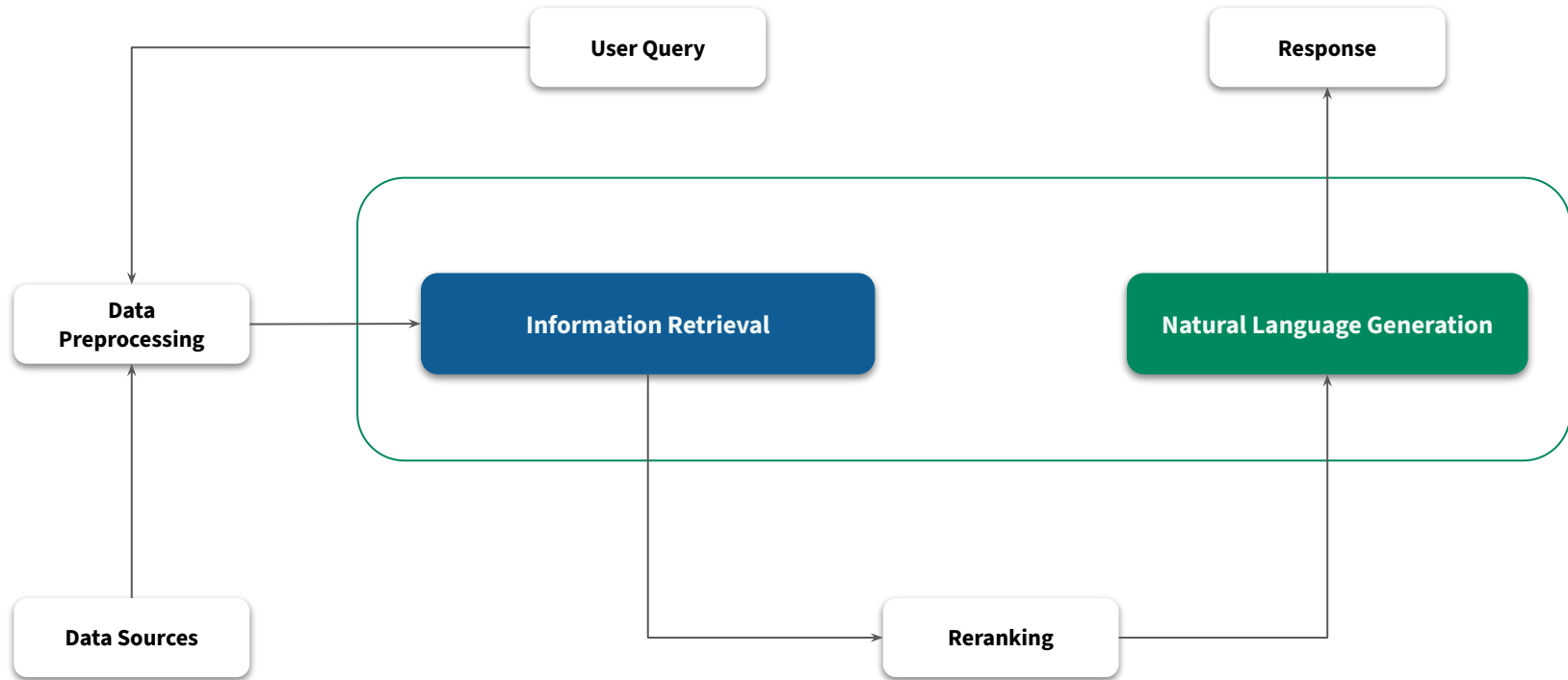


Agentic RAG



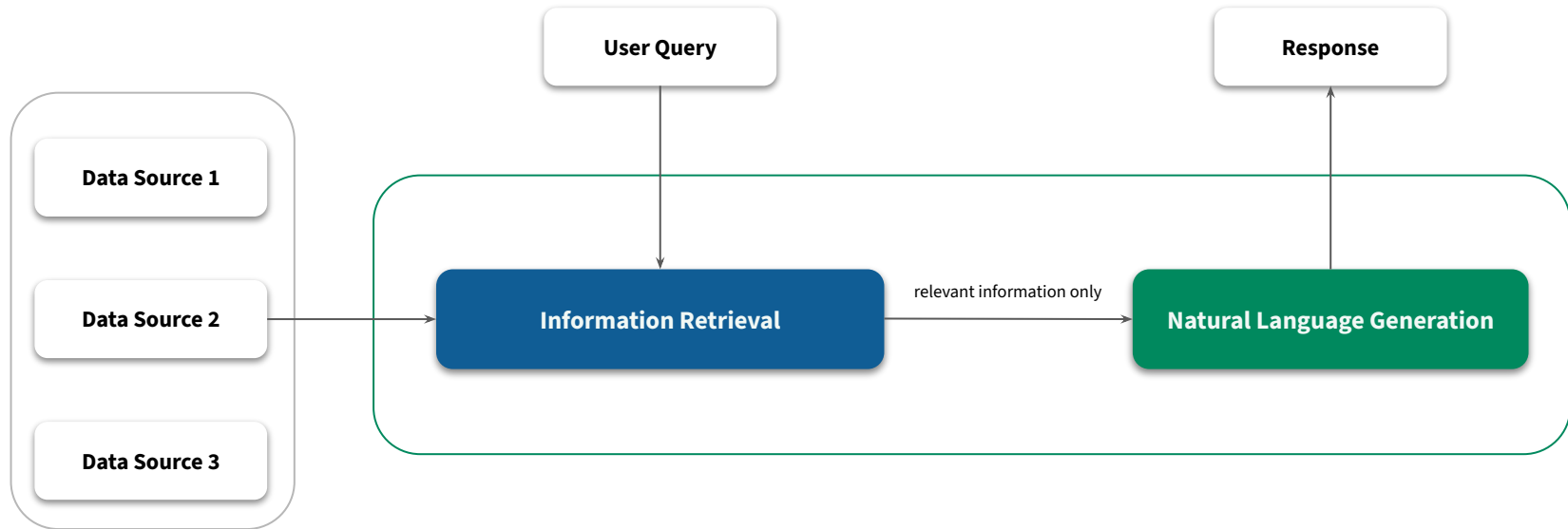


Advanced RAG



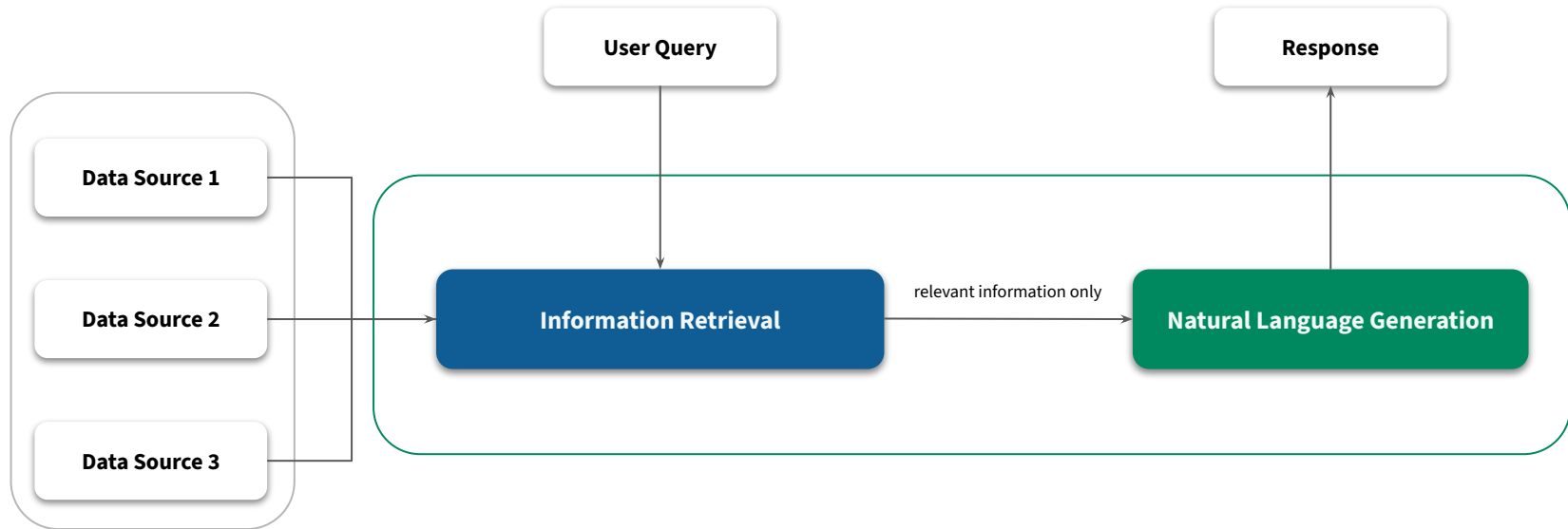


Branched RAG



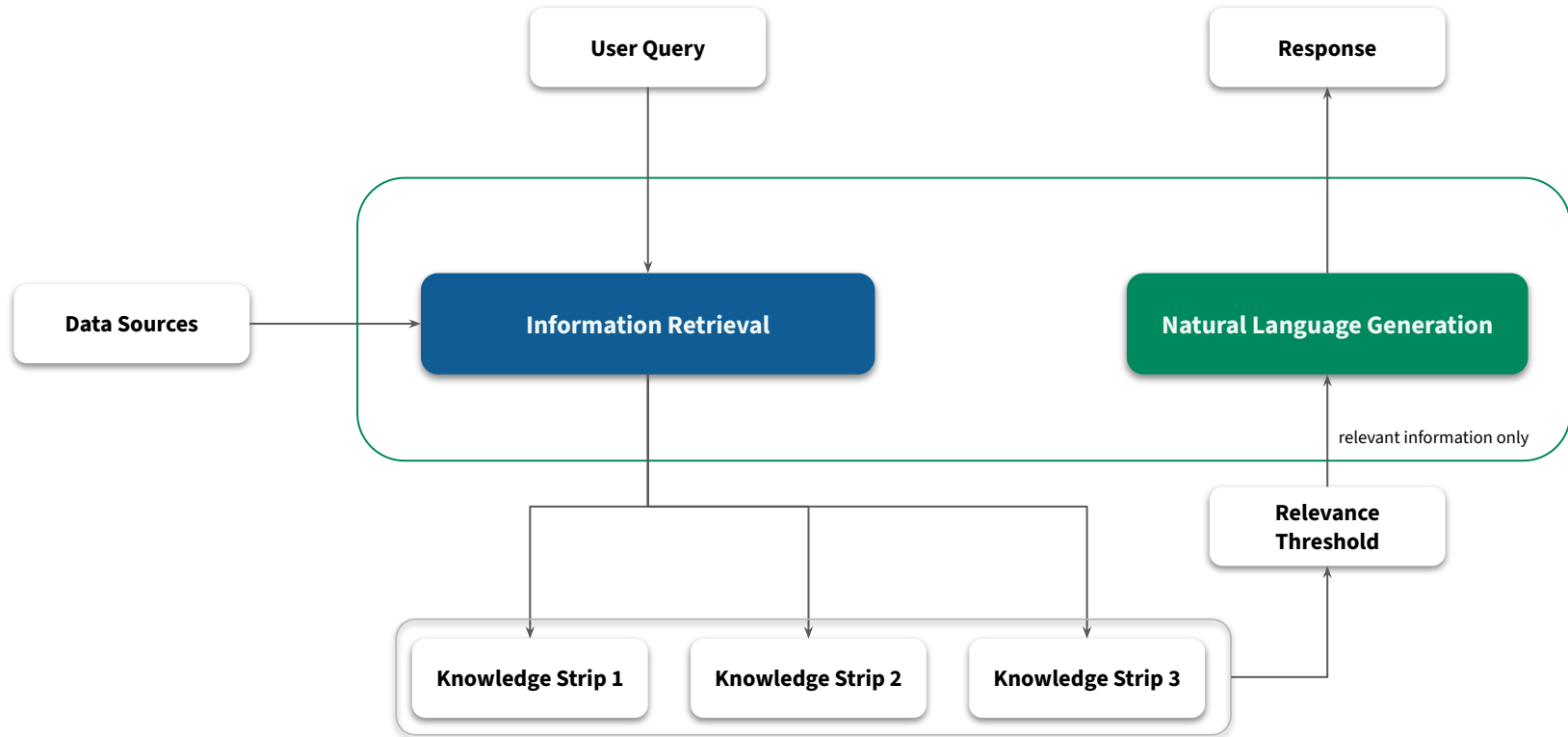


Adaptive RAG





Corrective RAG (C-RAG)





Aspects of RAG Implementation

What variation of RAG do you use?

What LLM do you use?

What Vector DB do you use?

What Embedding Model do you use?

What similarity metric do you use?

What chunking strategy do you use?